Master 2 Research internship proposal

A MATHEMATICAL MODEL FOR BACTERIAL SWIMMING IN BIOFILM MATRIX

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Biofilms are communities of bacteria embedded in a self-produced extracellular matrix, found in natural, medical and industrial environments [4]. Their formation is a recurring issue in medical settings, where they make bacteria more resistant to antimicrobials, but also in industrial settings, for example in sewers. To counterbalance these effects, adding swimming bacteria into the matrix can help diffuse chemicals in the whole bacterial community and irrigate it. Recent work by Ravel et al. [10] infers the characteristics of bacterial swimming in a biofilm matrix from time-lapse confocal laser scanning microscopy. Three bacterial species with different swimming speeds are considered, and a kinetic model of swimmer populations, including mechanistic interactions with the host biofilm, is inferred from $2D+T$ images. In that model, bacteria are treated as Lagrangian particles. This internship proposes to go one level below this population description: to model the swimmer itself, to recover its *true* trajectory depending on the species, and to confront the model with the results and data of [10].

Model. In [10], each bacterium is a Lagrangian particle whose position evolves according to a law on its velocity $\mathbf{v}$:

$$\mathrm{d}\mathbf{v} = \underbrace{\gamma(\alpha(b) - \|\mathbf{v}\|)\frac{\mathbf{v}}{\|\mathbf{v}\|}\mathrm{d}t}_{\text{speed selection}} + \underbrace{\beta\frac{\nabla b}{\|\nabla b\|}\mathrm{d}t}_{\text{direction selection}} + \underbrace{\eta \mathrm{d}t}_{\text{random term}},$$

where γ, β and η are scalars representing weights in front of each term and $\alpha(b)$ is a target velocity that depends on the host biofilm density b.

We wish to replace this effective description by a mechanistic single-swimmer model, drawing from [8]. There, the bacterium is described by Purcell's model [9] for *E. coli*: an ellipsoidal cell body with a rotating flagellum at one pole, immersed in a viscous fluid (see [7] for reviews of low-Reynolds-number swimming). Its motion is described by three quantities [8]: the swimming speed, ν, the flagellum angular speed, ω, and the body angular speed, Ω.

The general model parameters would need to be adapted to each species so that the simulated trajectory matches the observed one. The influence of the biofilm matrix would then then introduced either through the model parameters or through the structure of the fluid, and numerical experiments would be matched with data, e.g. from [6].

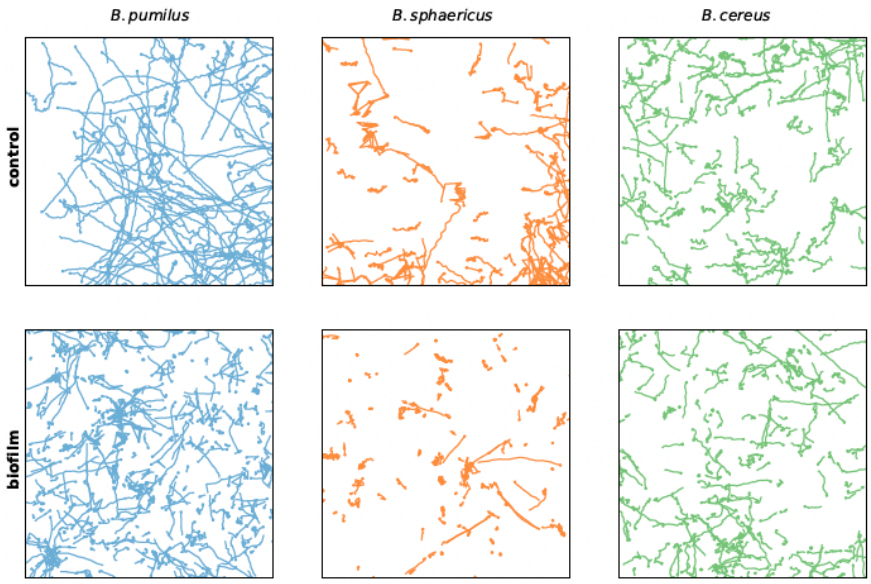


Figure 1: Swimmer trajectories in a Newtonian buffer (upper panel) and in a biofilm (lower panel). Three bacterial species are tested, with a number of trajectories $n = 517$ and 123 (*B. pumilus*), $n = 237$ and 94 (*B. sphaericus*) and $n = 279$ and 144 (*B. cereus*) for, respectively, the biofilm and the control buffer. Taken from [10].

Research directions. We have identified five main steps as listed below, focusing on a model for a single swimmer before extending it to a whole population:

- Understand and study Purcell's model [9] (see also [7]), first in Newtonian fluids.
- Review the literature on biofilm structure and study how a swimmer adapts to a biofilm matrix, comparing the data of [10] in a Newtonian fluid with those in a biofilm.
- Develop a comprehensive model for a single swimmer. Swimming strategies for flagellated organisms in viscous fluids could be inferred using [1], [2], and then adapted to biofilms by modifying fluid-structure interactions.
- Replace the Lagrangian particle model of [10] by the previously developed single-swimmer model in the equations for population dynamics.
- The dynamics of the previous models could be verified using particle tracking methods such as Euclidean random assignment problems (ERAPs) on synthetic data [3], and one could also infer model parameters using Physics-Informed Neural Networks [5].

Moreover, to go further, tests could then also be conducted on experimental data such as the one from [10]. From this, one could infer the minimum resolution and acquisition frequency of the data. Lastly, from the single-swimmer trajectory, infer the minimal number of swimmers and species (find an equilibrium between these two variables) needed to irrigate the biofilm matrix, as an optimization problem. A first question is whether this can be addressed by numerical methods alone (possibly finding lower and upper bounds on the two variables using ERAPs).

Keywords: micro-swimmers, partial differential equations, numerical analysis, biofilm community.

Candidate profile: You are a motivated Master 2 student in applied mathematics, with a solid background in ODEs, PDEs and numerical methods. Basic coding skills (especially in Python or MATLAB) are expected; a taste for modeling or an interest in biology is a plus; no prior knowledge of fluid mechanics or biofilms is required.

What we offer: The project can last **between 4** and **6 months** and is hosted by MaIAGE at INRAE Jouy-en-Josas, with a flexible starting date between **February and April 2027**. Close interaction with the MICALIS institute (R. Briandet, INRAE) could also give access to the experimental side of biofilm research. Regarding ERAPs, discussions with M. D'Achille (IECL, Université de Lorraine) and P. Moyal (LMV, Université Paris-Saclay) could also be possible. A pursuit as a PhD student on the subject can also be envisioned.

Application: We require a detailed CV (including a complete transcript of grades), a cover letter explaining your interest in the project, and the names of one or two permanent professors who are willing to write a recommendation letter. The application material should be sent via email to the two above emails before **December 31st 2026**.

Mentors **Jessie Levillain** is a research scientist at INRAE MaIAGE. She is specialized in mathematical modeling and numerical analysis, and works on swimmer models and on random assignment problems for particle tracking.

Lorenzo Sala is a research scientist at INRAE MaIAGE. He specializes in hybrid methods that combine data-driven and physics-based models to infer parameters in biological problems.

References

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